

Decomposing Lexicon-Assisted Correction for Word-Level Handwritten Text Recognition on IAM: Lexicon Coverage, Frequency Ranking and Line Context

Nur Banu Oğur*

*Department of Software Engineering, Faculty of Computer and Information Sciences,
Sakarya University, 54050 Sakarya, Türkiye
nbogur@sakarya.edu.tr*

Rıdvan Dursun

*Department of Software Engineering, Faculty of Computer and Information Sciences,
Sakarya University, 54050 Sakarya, Türkiye
ridvan.dursun@ogr.sakarya.edu.tr*

Berhat Yeşilyurt

*Department of Software Engineering, Faculty of Computer and Information Sciences,
Sakarya University, 54050 Sakarya, Türkiye
berhat.yesilyurt@ogr.sakarya.edu.tr*

Received (Day Month Year)

Revised (Day Month Year)

Accepted (Day Month Year)

Handwriting varies strongly between writers, which makes word-level recognition hard. We study it on the IAM database under the Aachen writer-disjoint protocol, on all 20,310 test words. Our system, CRNN-LX, is a convolutional-recurrent network trained from scratch, followed by a lexicon-aware post-corrector that rescores candidates with a Kneser–Ney trigram and decodes each text line jointly, conditioning every word on the recognizer’s own output for both of its neighbours. Two questions are examined. The first concerns the post-corrector, usually reported as a single language-model step. Decomposing it reveals four ingredients with different jobs: lexicon coverage decides the sign of the correction (a 7 K-word training vocabulary costs 2.4 points, a 239 K-type English list gains 0.7); the frequency prior that ranks candidates supplies most of the magnitude (1.2 points); an external corpus adds 0.6 and line context another 0.6, moving accuracy from 78.83% to 81.95% without raising the character error rate. The second question is whether writing-aware augmentation helps. Six configurations differing only in augmentation show no detectable gain from elastic, morphological and wide photometric transforms, whereas removing the conventional pipeline beneath them costs 2.74 points. Three seeds turn the remaining gap into noise: it changes sign between

*Corresponding author.

them.

Keywords: Handwritten text recognition; lexicon-based post-correction; Kneser–Ney trigram; line context; convolutional-recurrent neural network; connectionist temporal classification; data augmentation; ablation study.

1. Introduction

Offline *handwritten text recognition* (HTR) is the task of converting a scanned image of handwriting into machine-readable text without access to the pen trajectory. Depending on how the input is segmented, an HTR system may operate on full pages, on text lines, or — as in this work — on *word* images. Word-level recognition is the harder setting per sample, because the optical model sees one crop at a time and cannot exploit the surrounding words to disambiguate an unclear glyph. When the crops are cut from text lines, as in IAM, that context is not lost entirely: the recognizer’s own outputs for the neighbouring crops are available, and whether a post-corrector can use them is one of the questions examined here.

The IAM Handwriting Database¹⁴ is the standard benchmark for English HTR, containing roughly 115 000 word images from 657 writers. We adopt the partition released by RWTH Aachen University¹⁹ — referred to as the *Aachen split* throughout this paper — which guarantees that no writer contributes to more than one of the training, validation and test sets. This *writer-disjoint* protocol is stricter than a random split and is the realistic condition for deploying a recognizer on handwriting it has never seen.

Modern recognizers are trained with the *connectionist temporal classification* (CTC) objective,⁷ which removes the need for character-level alignment between the image and its transcription. Two model families dominate: convolutional-recurrent networks (CRNNs) trained with CTC,^{1,17,21} and attention-based sequence-to-sequence or transformer encoders.^{10,15} For a compact CRNN, two ingredients are commonly credited with closing part of the gap to heavier models: data augmentation that reflects how handwriting actually varies, and a lexical prior applied at or after decoding. Both are widely used, but their individual contribution is rarely measured under controlled conditions — same data, same architecture, same training schedule, same test set.

This paper provides such a measurement, and finds the two ingredients unequal. On the augmentation side, the object of study is three writing-aware transforms added on top of a conventional geometric and photometric pipeline (rotation, translation, scaling, shear, brightness and contrast jitter, Gaussian noise, gamma correction and random erasing; Section 3.4): *elastic deformation*, which imitates the small variations of a stroke from one writing to the next; *morphological perturbation*, which thins or thickens the strokes as a fine or a broad pen would; and a *widened photometric range*, which imitates lighter or darker ink and scans. Throughout the paper we call these three the *proposed transforms* and everything else the *conventional pipeline*. The label is ours rather than a standard: there is no agreed set of

routine augmentations for handwriting, and some prior systems already treat elastic distortion as one.⁵ On the lexical side, the post-corrector replaces a recognized word by a lexicon entry within a small edit distance and ranks the candidates with a word-level language model; in the proposed system that model is a Kneser–Ney trigram, and the words of a text line are decoded jointly, so that each choice is conditioned on the recognizer’s own output for the neighbouring crops on both sides. We name the proposed system CRNN-LX (CRNN with lexicon-extended, language-model-ranked post-correction) and its lexicon-free counterpart CRNN-G; Table 1 defines the six configurations compared. Our contributions are:

- **An anatomy of lexicon-assisted correction.** Holding the optical model fixed, we vary the post-correction lexicon, the language model that ranks the candidates, its training corpus and its context independently, which separates four effects that are usually reported as one. Coverage decides the sign: the training vocabulary alone *hurts*, by 2.4 to 2.7 percentage points (pp) on the five augmented models, while a general English list helps (+0.3 to +0.7 pp). A unigram frequency prior then supplies most of the magnitude (+1.1 to +1.3 pp), and its value grows with the lexicon, because a larger candidate set makes the ranking matter more. Replacing the prior by a trigram estimated on an external corpus adds 0.6 pp through the corpus and a further 0.6 pp through the line context — read from the recognizer’s own outputs, first from the left only and then from both sides by decoding the line jointly — and brings the character error rate back to that of lexicon-free decoding. The complete corrector delivers +3.1 pp, from 78.83% to 81.95%.
- **A controlled augmentation ablation with a positive control.** Six networks that differ only in the augmentation schedule are trained from scratch on the full Aachen training set and evaluated on the full test set. Elastic deformation, morphological perturbation and a widened photometric range change word accuracy by at most 0.32 pp, and none of those differences reaches the significance threshold ($p < 0.01$) used throughout; removing the conventional pipeline underneath them costs 2.74 pp at $p = 1 \times 10^{-30}$. Repeating the two endpoint configurations with three seeds each shows why: the difference between them changes sign across seeds (−0.49 to +0.78 pp) and the run-to-run spread of one configuration reaches 1.0 pp, several times the effect the transforms were expected to produce. These results are consistent with a saturation effect rather than with an inability of the experimental setup to detect augmentation-related improvements.
- **A methodological observation on elastic deformation.** The common parameterization — normalized Gaussian-smoothed noise scaled by $\alpha \in [2, 5]$ — yields root-mean-square (RMS) displacements of only 0.02–0.04 pixels per axis at the working resolution and is effectively a no-op; we report the corrected parameterization we used and its effect.
- **A verified evaluation protocol and full release.** Writer-, prompt- and form-level disjointness of the split are checked programmatically; all 20,310 test words

are used; code, split lists and per-word predictions are public.

Section 2 groups the related literature into four groups and positions our work within them. Section 3 describes the pipeline of Fig. 1. Section 4 formalizes the evaluation metrics and statistical tests. Section 5 reports the results, Section 6 analyzes them, and Section 7 concludes.

2. Related Work

We organize prior work into four groups according to how each relates to ours: (i) *recurrent CTC recognizers*, which share our optical backbone but are almost always evaluated on text lines; (ii) *attention and transformer recognizers*, which target the same word-level task as ours with a heavier decoding paradigm; (iii) *other word-level recognizers on IAM*, whose evaluation protocols differ from ours; and (iv) *lexical decoders*, which use a lexicon or language model inside the decoding loop, whereas our post-corrector applies it afterwards. For each group we state what we adopt and where we differ.

Recurrent CTC recognizers. Graves and Schmidhuber⁸ introduced multi-dimensional long short-term memory (LSTM) networks with CTC as an early end-to-end offline recognizer. Puigcerver¹⁷ showed that a one-dimensional bidirectional LSTM (BiLSTM) stacked on a convolutional neural network (CNN) feature extractor matches multi-dimensional recurrence at a fraction of the cost, defining the canonical CRNN baseline that our architecture follows. Bluche and Messina¹ added gated convolutions, and Flor et al.⁴ combined a CNN with a bidirectional gated recurrent unit (BGRU) and a language-model pipeline. Wigington et al.²⁴ studied augmentation for CNN-LSTM recognizers and reported gains from combined transforms on IAM lines. We adopt the CRNN backbone unchanged; our contribution lies in measuring, rather than assuming, the effect of the augmentation and of the lexical prior.

Attention and transformer recognizers. Sueiras et al.²³ and Kang et al.¹⁰ introduced sequence-to-sequence recognizers with content-based attention for isolated IAM words. Kang et al.⁹ later integrated a language model into the decoder itself through candidate fusion, and Kass and Vats¹¹ combined a residual-network (ResNet) encoder, attention decoding and transfer learning from scene-text corpora, reporting the strongest word-level accuracy on the Aachen split that we are aware of. These four systems share our task, metric and split and are listed in Section 5. Transformer architectures such as TrOCR¹³ have since lowered line-level error rates further, but require pre-training on hundreds of millions of synthetic lines. Multi-modal large language models have also been benchmarked on IAM;³ those systems operate on full pages and rely on proprietary pre-training corpora, so they are not comparable to a from-scratch word recognizer.

Other word-level recognizers on IAM. Dutta et al.⁵ improved a CNN–recurrent neural network (RNN) hybrid with synthetic pre-training, slant normalization and test-time augmentation and report a 12.61% unconstrained word error rate on IAM words evaluated without punctuation or letter case, a setting not on the same footing as a case-sensitive recognizer trained from scratch. Rajesh et al.¹⁸ re-trained that CNN–RNN together with their own HWRCNet, a CNN–BiLSTM operating in the JPEG discrete cosine transform (DCT) domain, on a random 95:5 partition of IAM and obtained 77.14% and 80.08% strict word accuracy on uncompressed images; we note that their headline figure of 89.05% is *word accuracy with flexibility* of two edit operations rather than strict accuracy. Taking a different route altogether, Mondal et al.¹⁶ recognize IAM words lexicon-free by detecting individual characters with a YOLOv3 object detector, trained on only 1 200 word images, and report a 29.21% word error rate on 30 000 randomly selected test words. All three systems use partitions of IAM other than ours, a point we return to below.

Lexical decoders. Most systems that exploit a vocabulary do so while decoding. Word beam search (WBS)²⁰ injects a dictionary constraint directly into CTC beam search, so that only lexicon words can be produced; candidate fusion⁹ feeds a language model’s proposals into an attention decoder; and Flor et al.⁴ couple the optical model to a language-model pipeline. In all of these the lexical knowledge is part of decoding and cannot be varied without re-decoding or retraining. We instead apply it *after* greedy decoding, which lets us swap the lexicon, the language model, its corpus and its context independently while the optical hypotheses stay fixed. That is the property on which the decomposition of Section 5.2 rests. The language model itself is standard: an interpolated Kneser–Ney n -gram,^{2,12} the usual smoothing for word n -gram models. What differs is where its context comes from: a line-level decoder reads the preceding words off the same image, whereas our corrector reads them off the recognizer’s outputs for the neighbouring word crops.

Positioning. Of the word-level systems above, only the attention-based recognizers of Sueiras et al.,²³ Kang et al.^{9,10} and Kass and Vats¹¹ are evaluated on the Aachen writer-disjoint partition that we use; the CRNN-family results of Rajesh et al.¹⁸ and the detector of Mondal et al.¹⁶ use their own random partitions of IAM. Comparisons across protocols are indicative only; the controlled comparisons of this paper are based on our own configurations.

Fig. 1 shows the complete processing chain, from the raw IAM word crops to the reported metrics, and marks the two stages whose effect this paper measures.

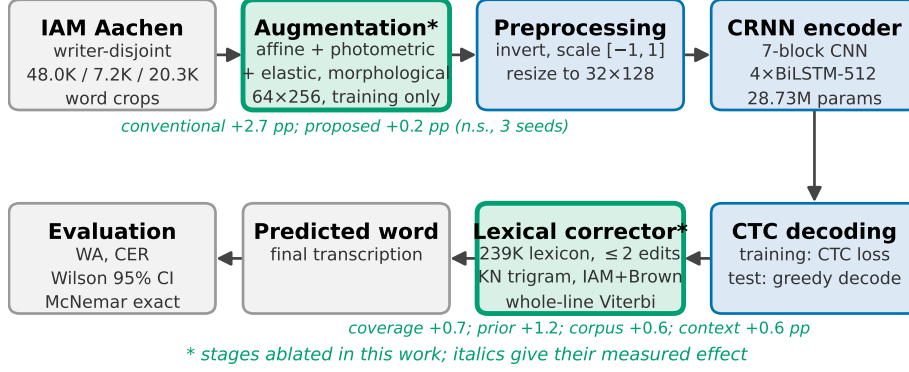


Fig. 1. Overview of CRNN-LX. Word crops are augmented (training only), normalized, encoded by a CRNN and decoded with CTC; the decoded string is then corrected against a 239 K-type lexicon, the candidates of all words on a text line being chosen jointly under a Kneser–Ney trigram, so that each word is conditioned on the system’s own output for its neighbours. The two outlined stages are the ones varied in the ablations, and each carries the effects measured for it. The evaluation stage reports word accuracy (WA), character error rate (CER) and Wilson confidence intervals (CI), all defined formally in Section 4.

3.1. Data and Protocol

We use the word-level IAM partition of the Aachen split:¹⁹ 747 training forms, 116 validation forms and 336 test forms with disjoint writer sets (283, 55 and 161 writers respectively). Following common practice, only word crops whose segmentation is flagged ok are retained. Five validation forms whose prompt text also occurs in the test set were removed from the validation partition so that checkpoint selection is independent of the test prompts; the test partition is used exactly as published. This yields 47 997 training words (two IAM image files are empty and are skipped), 7 205 validation words from 111 forms, and 20,310 test words from all 336 test forms. Writer, prompt and form disjointness between the three partitions, as well as the integrity of every record and image, are verified by a script distributed with the code. The label alphabet contains 78 symbols (upper- and lower-case Latin letters, digits and punctuation), plus the CTC blank.

3.2. Model Naming

Table 1 lists every configuration evaluated. All six share the optical model of Section 3.3, the training schedule and the post-corrector of Section 3.5; they differ only in the augmentation applied during training. CRNN-G is the fully augmented optical model decoded greedily, without any lexicon; CRNN-LX is that same model followed by the extended-lexicon post-corrector of Section 3.5 with the trigram and whole-line decoding, and is the configuration we propose. The suffixes denote the

post-processing rather than the augmentation, which turns out not to matter (Section 5.1). The first row is the zero-augmentation anchor: it has the conventional pipeline switched off as well, and exists so that the ablation can distinguish “the proposed transforms add nothing” from “this pipeline cannot detect augmentation at all”.

Table 1. Configurations compared. All use the same 28.73 M-parameter network, the same schedule and the same post-corrector. The rows are cumulative: elastic and morphological perturbation are each added on top of the widened photometric range. Conventional: the pipeline listed in Section 3.4.

Name	Conventional	Wide photometric	Elastic	Morphological
no augmentation	no	no	no	no
CRNN-B (baseline)	yes	no	no	no
+ wide photometric	yes	yes	no	no
+ elastic	yes	yes	yes	no
+ morphological	yes	yes	no	yes
CRNN-LX (proposed)	yes	yes	yes	yes

3.3. Optical Model

The encoder is a seven-block convolutional feature extractor that collapses the input to a single row of 512-dimensional frames, followed by a four-layer bidirectional LSTM with 512 hidden units and a linear projection onto the 79 output classes; the network has 28.73 M parameters. Inputs are grayscale crops resized to 32×128 , inverted so that ink is high-valued, and scaled to $[-1, 1]$. The network is trained with the CTC objective⁷ using AdamW with learning rate $\eta = 7 \times 10^{-4}$ and weight decay 10^{-5} , five warm-up epochs followed by cosine decay, mixed precision, and batches of 128, for at most 100 epochs. Training stops when neither the validation loss nor the validation word accuracy has improved for 15 epochs, and the checkpoint with the best validation word accuracy is evaluated once on the test set. The validation accuracy used for early stopping and checkpoint selection is computed with the extended lexicon and the unigram prior of Section 3.5, identically for all six configurations; the trigram corrector is applied only afterwards, to the frozen checkpoints, so that it never influences which weights are selected.

3.4. Augmentation

The baseline (CRNN-B) uses what we call the conventional pipeline: rotation ($\pm 7^\circ$), translation ($\pm 5\%$), scaling (0.9–1.1), shear ($\pm 5^\circ$), brightness and contrast jitter (0.85–1.15), Gaussian noise ($\sigma = 0.03$), gamma correction (0.80–1.20) and random erasing. The proposed transforms, which the ablation variants add cumulatively, are:

- **Wide photometric range.** Brightness and contrast 0.70–1.35, gamma 0.70–1.30, noise $\sigma = 0.05$.
- **Elastic deformation** (probability 0.5). A random displacement field is smoothed with a Gaussian kernel whose width is 8% of the larger image side, rescaled to unit RMS, and multiplied by an amplitude drawn uniformly from $[1, 3]$ pixels; the crop is warped accordingly. The unit-RMS normalization is essential: the frequently used variant that multiplies the *unnormalized* blurred noise by $\alpha \in [2, 5]^{22}$ produces, at the 64×256 working resolution, per-axis RMS displacements of only 0.02–0.04 pixels (never more than 0.19 px), i.e. it leaves the image visually and numerically unchanged. Our own earlier implementation had this property.
- **Morphological perturbation** (probability 0.15). The crop is eroded or dilated, with equal probability, by a 2×2 kernel, reproducing the difference between a fine and a broad nib.

Fig. 2 illustrates the transforms. All augmentation is applied on the graphics processing unit (GPU) to whole batches at a working resolution of 64×256 pixels (the median IAM word height is 68 px), with independent random parameters per sample, before the final resize to 32×128 ; the same code path is used by all six configurations. The zero-augmentation anchor bypasses that code path entirely, so its crops reach the network with preprocessing only.

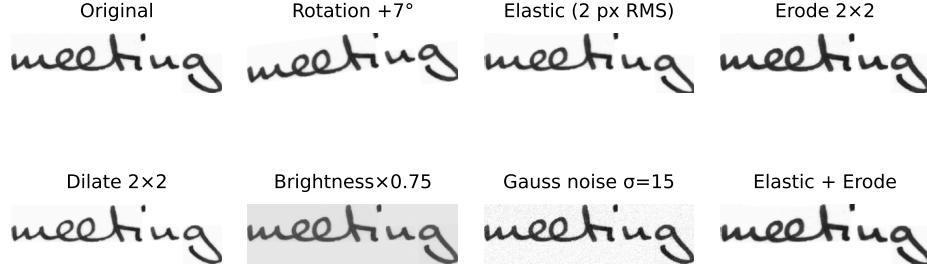


Fig. 2. Augmentation transforms on a training crop. The elastic panel uses the corrected amplitude (2 px RMS); the last panel shows the compound elastic+morphological effect.

3.5. *Lexicon-Extended Post-Correction with Line Context*

After greedy CTC decoding, each hypothesis is checked against a closed lexicon. If the hypothesis (case-insensitively) is in the lexicon it is accepted unchanged; otherwise it is replaced by the best-scoring lexicon entry within a Levenshtein distance of 1 (words of up to four characters) or 2 (longer words). On the test set 3 260 hypotheses (16.1%) fall outside the extended lexicon, and 2 939 of them have at least

one candidate within that distance; the other words pass through unchanged, so everything the corrector can gain or lose is decided among those 2 939. A candidate c at edit distance d from the hypothesis is scored as

$$s(c) = \log P(c \mid h_1, h_2) - \alpha d, \quad (1)$$

where P is a word-level language model, $h_1 h_2$ are the two words that precede the hypothesis on its text line, and α is an edit penalty selected on the validation partition. The lexicon, the language model and the way the line is decoded are the variables of interest, and Section 5.2 varies them one at a time with the optical model held fixed.

Lexicon. The *training lexicon* consists of the 7 173 word types of the training transcriptions; the *extended lexicon* adds the Natural Language Toolkit (NLTK) English word list, giving 239 126 types. Measured on the test transcriptions, the training lexicon covers 84.8% of the test tokens and the extended lexicon 93.9%; the remaining tokens are proper nouns, inflected forms absent from both lists, and punctuation-bearing strings.

Language model. Two models are compared. The *unigram frequency prior* ignores the context: it is the add-one-smoothed relative frequency of c among the 47 999 tokens of the training transcriptions, with $\alpha = 5$. It is the corrector of an earlier version of this work and serves as the reference. The *trigram with line context*, used by CRNN-LX, is an interpolated Kneser–Ney trigram² with absolute discount 0.75 and an additive floor over the lexicon vocabulary so that every candidate keeps a positive probability; an unseen trigram context degrades to the bigram and then to the unigram estimate. It is estimated on the same 47 999 tokens, read as text lines, and, in the proposed configuration, additionally on the Brown corpus⁶ (57 340 sentences, 1.16 M tokens). Tokens are case-sensitive and punctuation marks are kept as tokens, matching the IAM transcriptions.

Context. IAM word crops are cut from text lines and keep their position on the line, so the recognizer’s outputs for the neighbouring crops are available at test time. The context in Eq. (1) is the system’s *own output* for the neighbouring word crops of the same line — never their reference transcription. A line is a contiguous run of crops; the run is broken, and the context reset, wherever a crop is missing from the split (crops whose segmentation IAM flags as not **ok**). On the test set 88.5% of the words have a preceding crop. The optical model still sees one isolated crop at a time; only the post-corrector reads across the line, and the no-context figure is reported alongside for the setting in which words are genuinely isolated.

Decoding the line. Two decoders are compared. The *left-to-right* decoder corrects the words of a line in reading order, scoring each by Eq. (1) with $h_1 h_2$ set to its own two previous outputs, so a word is conditioned on its left neighbours

only. The *whole-line* decoder, used by CRNN-LX, instead chooses the word sequence $w_1 \dots w_n$ of the line that maximizes $\sum_i [\log P(w_i | w_{i-2}, w_{i-1}) - \alpha d_i]$ over all combinations of candidates, by exact second-order Viterbi search; since w_i also enters the terms of w_{i+1} and w_{i+2} , every choice is conditioned on the words to its left *and* right. Each out-of-lexicon hypothesis contributes a column of its candidates (at most the ten with the best context-free score); an in-lexicon hypothesis contributes only itself. Two options of the whole-line decoder are tested and selected on validation: whether an out-of-lexicon hypothesis may also stay as it is, as a candidate at distance zero with the model’s unseen-word floor probability, so that a correct word absent from the lexicon (a proper noun, an inflection) survives when the line supports it (*keep-OOV*); and whether an in-lexicon hypothesis may be replaced by a lexicon neighbour at distance one, so that real-word errors such as *form/from* become correctable (*real-word*). For the whole-line decoder α is selected from $\{1, 2, 3, 4, 5, 7, 10, 15, 20, 30\}$, a wider grid than the $\{3, 5, 7, 10\}$ of the left-to-right decoder because the line objective shifts the useful range. Every selection — α , the options, the decoder — is made on the validation writers and applied once to the test set.

External corpus. The IAM prompts were taken from the LOB corpus; Brown is a different corpus of the same design (American English, 1961), so overlap has to be measured rather than assumed. Of the word 5-grams on the test lines 0.63% occur verbatim in Brown, of the 6-grams 0.11%, of the 8-grams none; the longest shared sequence is seven words and the shared strings are idioms (*in spite of the fact that, on the other hand*). The same check against the IAM training lines gives 0.06% and 0%. The corpus therefore supplies English word statistics, not the test text, and the corrector never sees an IAM test transcription.

4. Experimental Setup

4.1. Evaluation Metrics

Let $\mathcal{T} = \{(\hat{y}_i, y_i)\}_{i=1}^N$ be the set of $N = 20,310$ predicted/reference word pairs. *Word accuracy* (WA) is the exact-match rate

$$\text{WA} = \frac{1}{N} \sum_{i=1}^N \mathbb{I}[\hat{y}_i = y_i], \quad (2)$$

and the word error rate (WER) follows as $\text{WER} = 1 - \text{WA}$. The *character error rate* (CER) aggregates Levenshtein distances $d(\cdot, \cdot)$ over the corpus,

$$\text{CER} = \frac{\sum_{i=1}^N d(\hat{y}_i, y_i)}{\sum_{i=1}^N |y_i|}, \quad (3)$$

so that a single badly recognized long word cannot dominate the score.

4.2. Confidence Intervals

Because WA is a proportion estimated on a finite test set, we report Wilson 95% confidence intervals (CIs) rather than normal-approximation intervals, which are unreliable near the boundaries. For k correct predictions out of N and $z = 1.96$,

$$\text{CI}_{95} = \frac{\hat{p} + \frac{z^2}{2N} \pm z \sqrt{\frac{\hat{p}(1-\hat{p})}{N} + \frac{z^2}{4N^2}}}{1 + \frac{z^2}{N}}, \quad \hat{p} = \frac{k}{N}. \quad (4)$$

With $N = 20,310$ the half-width of the interval is about 0.55 pp.

4.3. Significance Testing

Two systems evaluated on the same test set produce paired outcomes, so we use the exact McNemar test. Let b be the number of samples that the first system recognizes correctly and the second does not, and c the converse. Under the null hypothesis the discordant pairs are distributed $\text{Bin}(b + c, \frac{1}{2})$, giving the two-sided p -value

$$p = \min \left(1, 2 \sum_{i=0}^{\min(b,c)} \binom{b+c}{i} 2^{-(b+c)} \right). \quad (5)$$

We treat $p < 0.01$ as significant.

4.4. Reproducibility

All six configurations were trained on the same machine (one NVIDIA GeForce RTX 4070 Laptop GPU with 8 GB, PyTorch 2.7.1 with CUDA 11.8, Python 3.13) from the same code revision, with seed 42, at 35–75 s per epoch; a configuration takes roughly one hour. The two endpoint configurations, CRNN-B and CRNN-LX, were additionally re-trained with seeds 123 and 456 by one of the authors on a different machine (one NVIDIA T4 through a hosted notebook service), from the same code revision, the same split files — verified byte-identical apart from line endings — and the same hyperparameters, including the corrected elastic amplitude of Section 3.4, which we checked in the run logs of those trainings; only the seed and the hardware differ. The four intermediate configurations were trained once, and differences smaller than the confidence interval are not interpreted. Every number reported below comes from evaluation passes in single precision with cuDNN restricted to its deterministic algorithms, in which the six optical models and all post-corrections are scored against identical greedy hypotheses, model by model; two separate processes produce byte-identical hypotheses for all 20,310 test words. Without that restriction single precision was reproducible within a process but one word changed between processes, and mixed-precision inference shifted 2 words between runs (0.01 pp). The post-corrector is deterministic: candidates are enumerated in a fixed order (shorter first, then lexicographic), ties keep

the first candidate, and the output does not depend on the order in which lines are processed. Two numerical self-tests, distributed with the code, confirm that the trigram’s conditional distributions sum to one over the 271 303-type vocabulary for seen and unseen contexts, and that the whole-line decoder returns the same optimum as an exhaustive search over all candidate combinations on 400 random line layouts. Code, split definitions, the split-verification script, the language-model and leakage-check scripts, per-word predictions and evaluation scripts are available at <https://github.com/Ridvan013/CRNN-Handwriting-Recognition> (branch `feature/aachen-v3-extended-trigram`); the trained weights exceed the repository file-size limit and are distributed separately.

5. Results

5.1. Augmentation Ablation

Table 2 is the central result, and it separates into two layers.

The conventional pipeline matters. Removing it costs 2.74 pp (81.64 to 78.90%) and 1.69 pp of character error rate; 1 457 words are recognized only by the baseline against 900 only by the anchor, and the exact McNemar test rejects equality at $p = 1 \times 10^{-30}$. Without a lexicon the same comparison is wider still, 78.77 against 74.33% (−4.44 pp), so part of the weaker optical model is masked by the post-corrector.

The proposed transforms do not provide a statistically detectable improvement. The five configurations that keep the conventional pipeline reach 81.64–81.95%, and none differs from the baseline at the $p < 0.01$ threshold of Section 4. The full scheme (CRNN-LX) is nominally the best, 0.32 pp above the baseline ($p = 0.132$), which is within one confidence-interval half-width; elastic perturbation alone gives $p = 0.182$, morphological perturbation 0.466 and the widened photometric range 0.712. The comparison is also sensitive to the post-corrector, which is why we report it under all three: scored with the unigram prior the same two optical models differ by 0.37 pp at $p = 0.063$, with the left-to-right trigram by 0.44 pp at $p = 0.025$, and with the whole-line decoder by 0.32 pp at $p = 0.132$ — the same sign every time, never below the threshold.

Three seeds resolve the question, and they resolve it against the transforms. Table 3 repeats CRNN-B and CRNN-LX with seeds 123 and 456. The difference between the two changes sign: CRNN-LX is 0.32 pp above CRNN-B at seed 42, 0.49 pp *below* it at seed 123 and 0.78 pp above it at seed 456. Two of the three per-seed comparisons would be declared significant at the 0.05 level — in opposite directions ($p = 0.024$ and $p = 4 \times 10^{-4}$). Averaged over the three seeds the difference is $+0.20 \pm 0.64$ pp (paired $t(2) = 0.54$, $p = 0.64$), and one unchanged configuration spans a full point on its own (CRNN-LX: 81.43–82.44%). The run-to-run spread is therefore several times the effect under study, which is why we report the augmentation comparison as unresolved rather than as a small gain; the conventional pipeline, at 2.74 pp, sits an order of magnitude above that noise floor and is un-

touched by the argument. A practical consequence is that a single-seed McNemar test on 20K words can report a significant augmentation effect of either sign, so the per-configuration p -values of Table 2 should be read as descriptive. Because the configurations are nested, Table 2 reports two differences: Δ_B against the baseline CRNN-B, and Δ_P against the +wide photometric configuration on which elastic and morphological perturbation are built, which isolates each transform. By Δ_P , elastic perturbation adds 0.21 pp, morphological perturbation 0.07 pp and the two together 0.23 pp; the widened photometric range itself adds 0.08 pp, which is why Δ_B is larger by that amount (+0.29, +0.16 and +0.32 pp). On the per-word level these five disagree on 3 773 words (18.6%), yet the disagreements largely cancel: for the baseline–CRNN-LX pair, 841 words are recognized only by the baseline and 905 only by CRNN-LX. Among configurations that already augment, the choice of transform changes *which* words are recognized far more than *how many*.

Table 2. Augmentation ablation on the Aachen writer-disjoint test set ($N = 20,310$). All rows share the network, schedule and post-corrector (239K lexicon, Kneser–Ney trigram on IAM+Brown, whole-line decoding with keep-OOV; the edit penalty, selected on validation for each model separately, is $\alpha = 2$ or 3). Δ_B is the difference in WA to CRNN-B and Δ_P the difference to the +wide photometric configuration on which elastic and morphological perturbation are added; both are computed from unrounded accuracies. p is the exact McNemar test against CRNN-B on per-word outcomes. The first row is the zero-augmentation anchor and acts as a positive control. Best validation WA and the epoch at which it was reached are given for reference (validation: 7 205 words, 55 writers).

Configuration	WA (%)	CER (%)	Wilson 95% CI	Δ_B	Δ_P	p	best val WA (ep.)
no augmentation	78.90	10.52	[78.33, 79.45]	−2.74	—	1×10^{-30}	82.57 (38)
CRNN-B (baseline)	81.64	8.83	[81.10, 82.17]	(ref.)	—	(ref.)	84.61 (87)
+ wide photometric	81.72	8.39	[81.19, 82.25]	+0.08	(ref.)	0.712	84.34 (76)
+ elastic	81.93	8.21	[81.39, 82.45]	+0.29	+0.21	0.182	84.69 (77)
+ morphological	81.80	8.25	[81.26, 82.32]	+0.16	+0.07	0.466	84.58 (84)
CRNN-LX (all)	81.95	8.54	[81.42, 82.48]	+0.32	+0.23	0.132	84.77 (70)

Table 3. Seed repeats of the two endpoint configurations, scored with the corrector of Table 2 ($N = 20,310$). Seed 42 is the run reported throughout the paper; seeds 123 and 456 differ from it only in the seed and in the machine they were trained on (Section 4). Δ is CRNN-LX minus CRNN-B within a seed and p the exact McNemar test on that pair; the two differences that reach $p < 0.05$ have opposite signs.

Seed	CRNN-B WA (%)	CRNN-LX WA (%)	Δ (pp)	p	CER: CRNN-B / CRNN-LX (%)
42	81.64	81.95	+0.32	0.132	8.83 / 8.54
123	81.93	81.43	−0.49	0.024	8.20 / 8.91
456	81.66	82.44	+0.78	4×10^{-4}	8.53 / 8.28
mean $\pm$ SD	81.74 $\pm$ 0.16	81.94 $\pm$ 0.50	+0.20 $\pm$ 0.64	0.644 ^a	8.52 $\pm$ 0.32 / 8.57 $\pm$ 0.32

^aPaired t test on the three per-seed differences, $t(2) = 0.54$; the p values above it are per-seed McNemar tests.

Fig. 3 shows the mechanism. The anchor separates from the rest early: by epoch

53 its training loss is down to 0.018, where the augmented runs are still at 0.056–0.070, while its validation accuracy peaks at 82.57% at epoch 38 and early stopping ends the run at epoch 53. It fits the training set far more tightly and generalizes worse: its validation loss ends at 0.46 against 0.32–0.37 for the augmented runs. Those five keep improving to epoch 70–87 and converge to a plateau of 84.3–84.8% validation accuracy, and among them the curves are indistinguishable after the first fifteen epochs; their training losses differ a little (the fully augmented model fits the training set least tightly, as a regularizer should), but the spread that reaches validation accuracy is only 0.4 pp. With 47 997 training words from 283 writers, the conventional pipeline already covers the variability that the proposed transforms simulate.

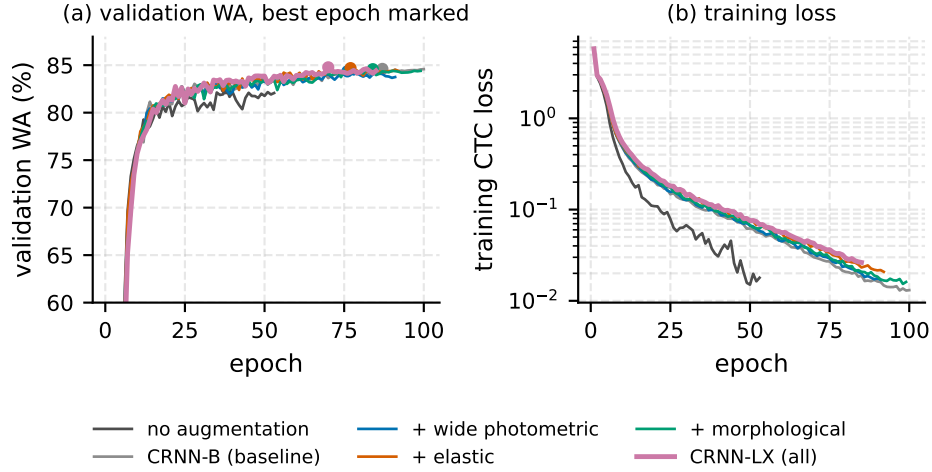


Fig. 3. Training dynamics of the six configurations: (a) validation word accuracy with the best epoch marked, (b) training CTC loss. The zero-augmentation anchor drops fastest in training loss, stops earliest and reaches the lowest validation accuracy; the five augmented curves coincide.

5.2. *Lexicon Ablation*

Table 4 holds the optical model fixed (CRNN-LX, best checkpoint) and varies only what happens after greedy decoding. The effect is an order of magnitude larger than that of the augmentation. The corrector has four ingredients — the lexicon a hypothesis is tested against, the language model that ranks the candidates within edit distance, the corpus that model is estimated on, and the context it is given — and the table varies them one at a time, so that rows 2 and 4 isolate coverage at fixed ranking, rows 2→3 and 4→5 isolate the frequency prior at a fixed lexicon,

rows 5–9 isolate the corpus and the left context at a fixed lexicon, and rows 9–13 vary only how the line is decoded.

Coverage decides the sign. Against the training vocabulary alone, accuracy *falls* from 78.83% to 76.42%: hypotheses that are valid English words absent from the 7 K training list — 15% of the test tokens — are forced onto the nearest training word, and most of these substitutions are wrong. Replacing that list with 239 K types, which covers 94% of the test tokens, turns the same -2.42 pp into $+0.67$ pp (76.42 to 79.51%, a swing of 3.09 pp at identical ranking).

The frequency prior supplies most of the magnitude, and its value grows with the lexicon. On the 7 K list it is worth $+0.63$ pp (76.42 to 77.04%); on the 239 K list, $+1.23$ pp (79.51 to 80.74%). Fig. 4(a) shows both effects at once. The reason is combinatorial rather than linguistic: a larger lexicon puts more entries within the same edit distance of a hypothesis, so the choice among equidistant candidates — which is exactly what the prior makes — decides more cases, and Section 6.2 takes the argument further. Neither ingredient alone reaches the $+1.91$ pp of the unigram corrector.

The corpus and the context supply the rest (Fig. 4(b)). Rows 5–9 keep the extended lexicon and change only the language model of Eq. (1). Replacing the add-one prior by a Kneser–Ney unigram estimated on the same IAM text changes nothing (80.74% in both cases; 19 words gained, 18 lost), so the smoothing is immaterial. Giving that model the line context, still on IAM text alone, adds only 0.10 pp (80.84%, $p = 0.022$): 48 K tokens of training text rarely contain the context of a test word. Adding the Brown corpus to the context-free model adds $+0.59$ pp (81.32%, $p = 9 \times 10^{-19}$; 155 words gained against 36 lost), and the trigram with line context on the enlarged corpus adds a further $+0.34$ pp, reaching 81.66% ($p = 1 \times 10^{-35}$ against the unigram prior; 220 words gained against 32 lost). Decomposed in the other order — context first, corpus second — the shares are $+0.10$ and $+0.83$ pp, so the two ingredients are not additive: context is worth little until the corpus is large enough to have seen it. Two diagnostics bound this step. Replacing the system’s own outputs by the reference transcriptions of the neighbouring crops (an oracle, not a reportable system) gives 81.73%, only 0.06 pp more, so almost nothing is lost to the recognizer’s own errors in the context. And the edit penalty is not what carries the result: $\alpha = 7$ was selected on the validation set, but every value tried (3, 5, 7, 10) gives between 81.43 and 81.66% on the test set, all above the unigram prior.

Decoding the line jointly adds the last step (rows 10–13). With the same trigram, choosing all words of a line together instead of left to right — so that a word is also conditioned on what follows it — gains 0.11 pp (81.77%, $p = 0.026$; 56 words gained, 34 lost). Allowing an out-of-lexicon hypothesis to survive as a candidate (keep-OOV) gains a further 0.18 pp, reaching 81.95% ($p = 0.032$ against the left-to-right decoder, 394 gained against 335 lost), and it does so by making the corrector *more* conservative: it now replaces 1925 hypotheses instead of 2939, repairs 815 and breaks 181, where the left-to-right decoder repaired 1076 and broke 501. Al-

lowing real-word substitutions (row 12) reaches 81.94% on the test set with fewer but surer changes (141 gained, 86 lost, $p = 3 \times 10^{-4}$), but it scored below keep-OOV on the validation set and was therefore not selected; the two options together (row 13) interact badly, since a hypothesis that keeps itself at distance zero now competes with real-word substitutions of its neighbours, and lose 2.2 pp. The whole-line decoder with keep-OOV was selected on validation and is the configuration of CRNN-LX; on the four other augmented models of Table 2 it gains 0.32–0.44 pp over the left-to-right decoder ($p = 0.002$ – 0.034), and 0.08 pp ($p = 0.57$) on the zero-augmentation anchor, whose weaker hypotheses give the line objective less to work with. In all, the corrector moves CRNN-G from 78.83% to 81.95%: coverage +0.7, frequency prior +1.2, corpus +0.6, line context +0.6 pp. The unigram pattern holds on the other four augmented optical models of Table 2: the training lexicon costs 2.4–2.7 pp, the extended lexicon gains 0.3–0.6 pp over lexicon-free decoding, the prior adds a further 1.1–1.3 pp, and the complete corrector gains 1.5–1.9 pp. On the zero-augmentation anchor, whose greedy output is 4.4 pp weaker, the ordering is unchanged but every effect is larger and the complete corrector gains 3.54 pp: the more the optical model errs, the more a lexical prior has to repair.

Table 4. Post-correction ablation with the optical model fixed (CRNN-LX checkpoint, $N = 20,310$). Each row applies a different post-correction to the *same* greedy hypotheses; lexicon, language model, corpus, context and line decoding vary one at a time. KN1/KN3: interpolated Kneser–Ney unigram/trigram; L→R: the trigram is given the system’s own outputs for the two preceding crops of the line; line: whole-line (Viterbi) decoding, conditioned on both neighbours. Edit penalty $\alpha = 5$ for the add-one prior; otherwise selected on the validation set (5 for row 6, 7 for rows 7–9, 10 for rows 10, 12 and 13, 2 for row 11). Rows 12 and 13 are reported for completeness; they scored below row 11 on validation and were not selected.

Post-correction	Ranking (corpus)	Lexicon	WA (%)	CER (%)	Wilson 95% CI
none (CRNN-G, greedy CTC)	—	n/a	78.83	8.60	[78.27, 79.39]
training lexicon, edit distance only	—	7 173	76.42	11.19	[75.83, 76.99]
training lexicon + frequency prior	add-one unigram (IAM)	7 173	77.04	10.94	[76.46, 77.61]
extended lexicon, edit distance only	—	239 126	79.51	9.71	[78.95, 80.06]
extended lexicon + frequency prior	add-one unigram (IAM)	239 126	80.74	9.09	[80.19, 81.28]
extended lexicon + KN1	KN1 (IAM)	239 126	80.74	9.09	[80.20, 81.28]
extended lexicon + KN3, L→R	KN3 (IAM)	239 126	80.84	9.07	[80.29, 81.37]
extended lexicon + KN1	KN1 (IAM+Brown)	239 126	81.32	8.74	[80.78, 81.85]
extended lexicon + KN3, L→R	KN3 (IAM+Brown)	239 126	81.66	8.58	[81.13, 82.19]
extended lexicon + KN3, line	KN3 (IAM+Brown)	239 126	81.77	8.54	[81.24, 82.30]
extended lexicon + KN3, line, keep-OOV (CRNN-LX)	KN3 (IAM+Brown)	239 126	81.95	8.54	[81.42, 82.48]
extended lexicon + KN3, line, real-word	KN3 (IAM+Brown)	239 126	81.94	8.51	[81.40, 82.46]
extended lexicon + KN3, line, keep-OOV + real-word	KN3 (IAM+Brown)	239 126	79.71	8.42	[79.16, 80.26]

The character error rate does not follow word accuracy. A whole-word corrector can only replace a hypothesis by a complete lexicon entry; when the replacement is wrong, a near-miss with one character error becomes a different word with several. The training lexicon, which replaces many correct-but-unlisted words, pays this price most heavily (11.19% against 8.60% lexicon-free); the extended lexicon with the unigram prior pays it less (9.09%) but still raises CER by half a point while

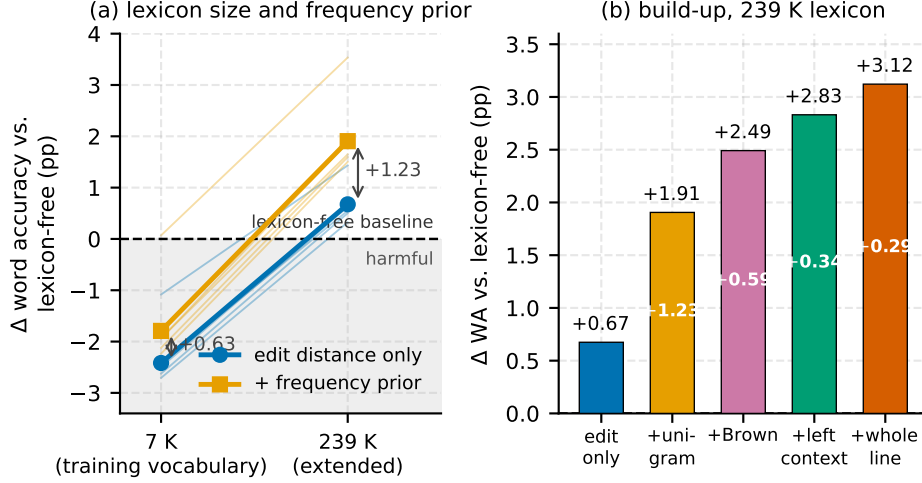


Fig. 4. What the post-corrector contributes, measured against lexicon-free decoding. (a) Coverage decides the sign: with 7 K types both variants fall below the baseline, with 239 K both rise above it. The vertical gap between the two lines is the contribution of the frequency prior, and it doubles as the lexicon grows. Thick lines: the CRNN-LX optical model; faint lines: the other five configurations of Table 2. (b) Build-up on the 239 K lexicon: edit distance alone, the unigram prior, the Brown corpus, left-to-right line context, and whole-line decoding with keep-OOV (white labels: increment over the previous bar).

improving WA by almost two. Only the trigram on the larger corpus chooses the right candidate often enough to bring CER back to the lexicon-free level (8.58% left to right, 8.54% with whole-line decoding). Lexicon-assisted word accuracy and character error rate therefore measure different things and should be reported together.

5.3. Comparison with Prior Word-Level Systems

Table 5 places CRNN-LX next to seven published word-level systems and states, for each, whether a lexicon or language model was applied at test time. Three of our own rows are given, because they answer different questions: CRNN-G is the optical model alone; the *no-context* row applies the extended lexicon with the unigram prior and is the figure for words that are genuinely isolated; CRNN-LX additionally reads the recognizer’s own outputs for the neighbouring crops of the line, which none of the listed systems does. The upper block lists results obtained on the Aachen writer-disjoint partition. The closest match in protocol is the sequence-to-sequence recognizer of Sueiras et al.:²³ the sizes of their partition (47 952, 20 306 and 7 558 words, as compiled in¹⁶) are those of the ok-filtered Aachen training, test and validation lists (47 999, 20 310 and 7 559) from which ours are drawn, and they report lexicon-free word and character error rates on the full test partition

exactly as we do. Under lexicon-free decoding CRNN-G is 2.6 pp more accurate (78.83% against 76.20%) at a slightly lower CER (8.60% against 8.80%); with the extended lexicon the margin is 4.5 pp without context and 5.8 pp with it. CRNN-LX remains below the later attention-based systems: 0.6 pp below Kang et al.,¹⁰ 2.1 pp below the candidate-fusion model of Kang et al.⁹ and 2.7 pp below AttentionHTR,¹¹ which additionally transfers knowledge from 14.4 M scene-text images; without line context each gap is 1.2 pp wider (1.8, 3.4 and 3.9 pp), and against our lexicon-free figure 3.1 pp wider. The gap is larger at character level (5.79–6.88% against our 8.54–8.60%), consistent with the autoregressive decoders of those systems modeling inter-character dependencies that a CTC output with whole-word post-correction cannot. The lower block reports three systems measured on other partitions of IAM; CRNN-LX is above all three and CRNN-G above two, but the partitions differ in writer overlap and size and we draw no conclusion from margins of this size.

Table 5. Word-level results on IAM. WA in parentheses is derived as $1 - \text{WER}$ from the cited paper. Lexicon: lexical constraint at test time (none = unconstrained decoding; LM = language model without a closed lexicon; n/s = not stated). Only CRNN-LX uses the recognizer’s outputs for the neighbouring crops of the line; every other row recognizes each word in isolation.

System	Year	Lexicon	WA (%)	CER (%)
<i>Aachen writer-disjoint partition (≈ 20.3 K test words)</i>				
Sueiras et al. ²³	2018	none	(76.20)	8.80
Kang et al. ¹⁰	2018	none	(82.55)	6.88
Kang et al. ⁹	2021	LM	(84.09)	5.79
Kass and Vats ¹¹	2022	none	(84.60)	6.50
CRNN-G (ours)	2026	none	78.83	8.60
CRNN-LX, no context (ours)	2026	239 K + unigram prior	80.74	9.09
CRNN-LX (ours)	2026	239 K + trigram, line context	81.95	8.54
<i>Other partitions of IAM</i>				
Mondal et al. ^{16a}	2022	none	(70.79)	9.53
CNN-RNN of, ⁵ re-trained in ^{18b}	2022	n/s	(77.14)	11.08
Rajesh et al. ^{18b}	2022	n/s	80.08	9.89

^a30 K randomly selected test words; 1 200 training images.

^bRandom 95:5 split, writer-dependent, ≈ 5.8 K test words.

5.4. Error Analysis

CRNN-LX misrecognizes 3 665 of the 20,310 test words. Only 275 of them (7.5%) differ from the reference in letter case alone; the remaining errors are genuine. At character level the aligned errors consist of 5 219 substitutions, 1 286 deletions and 486 insertions, so the model drops characters far more often than it hallucinates them. Fig. 5 shows the ten most frequent substitutions: they are mostly symmetric confusions between visually similar lower-case glyphs — $\text{o} \leftrightarrow \text{a}$ (257 cases in both directions), $\text{s} \leftrightarrow \text{r}$ (151), $\text{n} \leftrightarrow \text{m}$ (107), $\text{t} \leftrightarrow \text{l}$ (101), $\text{n} \leftrightarrow \text{r}$ (90), $\text{e} \leftrightarrow \text{a}$ (87). Case substitutions account for 7.8% of all substitutions. Of the post-corrector’s decisions,

1925 hypotheses (9.5%) are replaced: 815 wrong hypotheses become correct and 181 correct ones, valid words absent from the lexicon, become wrong. The left-to-right decoder without keep-OOV replaces 2939, repairs 1076 and breaks 501; the unigram prior replaces the same 2939 and breaks the same 501 — the lexicon decides which words are touched — but repairs only 888. Whole-line decoding with keep-OOV thus wins not by repairing more but by leaving alone the words it would have broken. The confusions themselves are errors of shape, not of vocabulary: a whole-word lexicon can repair them only when the mis-shaped word falls outside the lexicon and its correct form lies within two edits, which is exactly the population that the extended lexicon recovers in Table 4.

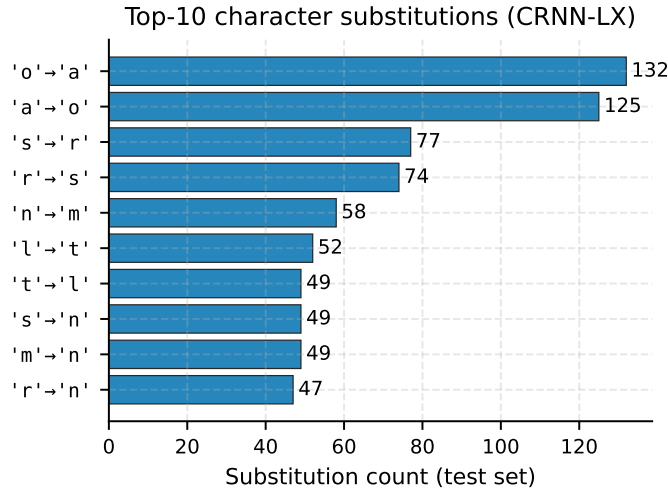


Fig. 5. Ten most frequent character substitutions of CRNN-LX on the test set (true → predicted).

6. Discussion

6.1. Interpretation of the Augmentation Results

The anchor rules out the explanation that would otherwise be hardest to exclude. A null result can always mean that the experiment is too noisy to see anything, and the seed repeats confirm that the noise is real: an unchanged configuration moves by a full point from seed to seed. But the same test, the same data and the same protocol detect the removal of the conventional pipeline at $p = 1 \times 10^{-30}$, 2.7 points of word accuracy. The apparatus is not blind to augmentation; it cannot resolve *these* transforms on *this* training set, whose largest effect (+0.32 pp, $p = 0.132$) stays below the threshold, changes with the post-corrector (+0.44 pp, $p = 0.025$ under the left-to-right corrector) and changes sign with the seed (Table 3). What

we observe is consistent with saturation, and three further observations indicate where it occurs.

First, the anchor overfits earliest and hardest: at epoch 53, where it stops, its training loss is 0.018 against 0.056–0.070 for the augmented runs, and its validation loss has risen 0.10 above its own minimum, against 0.03–0.07 for them; the augmented runs stop 32 to 47 epochs later (Fig. 3). The first increment of augmentation buys regularization that the network genuinely needs. Second, among the augmented configurations the validation curves coincide and the extra transforms neither slow convergence nor raise the plateau, while the fully augmented model reaches a slightly higher training loss — so they do regularize further, but the baseline is no longer overfitting in a way that costs accuracy on unseen writers. Third, the per-word disagreements between those configurations are large yet balanced. Together these indicate that with 48 K training words from 283 writers, the writer-to-writer variation already present in the data exceeds what elastic and morphological perturbation of 64-pixel-high crops can add on top of affine and photometric jitter. Reports of sizeable gains from such augmentation, including our own preliminary experiments, came from smaller training sets or from comparisons in which augmentation was not the only difference. The result does not imply that augmentation is useless for HTR in general — Wigington et al.²⁴ report gains on lines with a different transform family, and our own anchor shows the first increment is worth 2.7 pp — but it does show that further transforms must be measured under controlled conditions before being claimed.

6.2. Coverage and the Frequency Prior

The ablation separates four effects that are usually conflated under “adding a language model”, and they turn out to do different jobs. Membership testing against a list that covers 85% of the test tokens is harmful, because the corrector cannot distinguish a misrecognized word from a correctly recognized word it has never seen and treats both as errors; raising coverage to 94% is what turns the loss into a gain. That gain is small on its own, however. What converts it into +1.9 pp is the prior that ranks the candidates, and it is worth twice as much on the large lexicon as on the small one.

The unigram prior is not doing linguistic work; it breaks ties. The tighter the lexicon, the fewer entries lie within two edits of a hypothesis and the less the ranking matters; at 239 K types a mis-shaped word typically has several equidistant neighbours, and picking the frequent one is right more often than picking an arbitrary one. Edit distance identifies the candidate set and frequency chooses inside it: reporting either without the other describes only half of the corrector.

The trigram does linguistic work, but only once it has text to learn it from. On the 48 K tokens of IAM training text the line context is worth 0.10 pp, because the two words preceding a test word rarely form a context the model has seen; on 1.2 M tokens it is worth 0.34 pp, and the corpus itself, even without context, is

worth 0.59 pp, because 7 K types observed once or twice give a poor estimate of which of two candidates is the commoner English word. Decoding the line jointly adds another 0.29 pp, but the larger part of that comes from a different mechanism than context: letting an unlisted hypothesis survive turns the corrector from one that must replace every unlisted word into one that may, and the line objective then decides. The left-to-right corrector broke 501 correctly recognized words that the lexicon did not contain; the whole-line decoder breaks 181 (Section 5.4). The ceiling is close: the corrector only ever touches the 16% of hypotheses that fall outside the lexicon, and with reference transcriptions as context the left-to-right corrector gains a further 0.06 pp only. What remains after correction is therefore not a ranking problem. The character error rate confirms the division of labour: the unigram corrector buys word accuracy at the price of character accuracy (9.09% against 8.60% lexicon-free), because a wrong whole-word substitution turns a near miss into several character errors. The trigram replaces exactly the same hypotheses but picks the right candidate for 1 076 of them instead of 888, and ends at the lexicon-free value (8.58% left to right, 8.54% whole-line).

A practical consequence is that word-level HTR results obtained with a closed lexicon should state the lexicon’s coverage of the test vocabulary, the rule used to rank candidates, the corpus the ranking was learned from and whether any context was used, without which the reported accuracy is not interpretable. The remaining 6% of uncovered tokens (proper nouns, rare inflections, punctuation-bearing strings) bound what any whole-word corrector can achieve; addressing them requires an open-vocabulary character-level model.

6.3. Comparison with Previous Word-Level HTR Systems

Among the systems of Table 5, the recognizer of Sueiras et al.²³ is the one whose protocol matches ours most closely: isolated IAM words, the Aachen writer-disjoint partition drawn from the same ok-filtered word lists — the only prior system for which we could verify this from the reported partition sizes (Section 5.3); we additionally discard two unreadable images and five validation forms whose prompts recur in the test set — and lexicon-free word and character error rates on the full test partition. Architecturally the two differ in every stage: they extract features from a sliding window of image patches with a LeNet-5-style convolutional network and transcribe them with a recurrent encoder–decoder using attention, whereas CRNN-G is a single convolutional-recurrent network trained with CTC. Under lexicon-free decoding CRNN-G reaches 78.83% against their 76.20% (23.8% word error rate), a margin of 2.6 pp that is almost five times the half-width of our confidence interval, at a slightly lower character error rate (8.60% against 8.80%). Post-correction with the extended lexicon widens the word-level margin to 4.5 pp without context and 5.8 pp with it. Sueiras et al. also report lexicon-constrained variants of their recognizer; those depend on the lexicon employed and are not compared here. We read the comparison as evidence that a plain CRNN trained from

scratch on the 48K Aachen training words is competitive with an early attention model under the same protocol, not as evidence about augmentation: the proposed transforms contribute nothing measurable to that margin (Tables 2 and 3).

HWRCNet,¹⁸ the closest system in construction (a CNN, a bidirectional LSTM and a CTC layer), reports 80.08% strict word accuracy, 1.87 pp below CRNN-LX (0.66 pp below its no-context variant) and 1.25 pp above CRNN-G. It is measured on a random 95:5 partition whose test set is writer-dependent and about 3.5 times smaller than ours, so neither difference is interpretable.

6.4. *Threats to Validity*

The augmentation differences of 0.1–0.3 pp are below what one training run can resolve: with three seeds the CRNN-B–CRNN-LX difference changes sign and averages $+0.20 \pm 0.64$ pp (Table 3), so we report the transforms as unresolved rather than as small gains. Only the two endpoint configurations were repeated; the four intermediate ones were trained once, and their individual p -values in Table 2 therefore carry no more weight than the seed-42 row of Table 3 does. The repeats also ran on different hardware, so the spread we quote includes any hardware-related nondeterminism — which only reinforces the conclusion that an effect of this size is not resolvable here. The 2.74 pp effect of the conventional pipeline is an order of magnitude larger than that resolution limit, so the positive control does not depend on the seed in the way the null result does. The lexicon and ranking effects (−2.4 to −2.7 pp for the training vocabulary, +1.1 to +1.3 pp for the frequency prior) are reproduced in sign and ordering on all six independently trained models and are well outside the interval, although their magnitude grows as the optical model weakens. The gain of the left-to-right trigram over the unigram prior is equally stable: +0.85 to +0.96 pp on each of the six optical models, with $\alpha = 7$ selected on validation for all of them; whole-line decoding then adds +0.29 to +0.44 pp on the five augmented models and +0.08 pp on the anchor. The whole-line decoder has two options and a wider penalty grid, all selected on the 7 205 validation words; a variant that was not selected (real-word substitution) reaches almost the same test accuracy by a different route, which shows that the selection is not finely tuned to the test set but also that the validation set cannot separate the two. These effects are, however, measured with one candidate generator and one word list; a different edit-distance bound or a different general vocabulary would change their size, though the ordering of the post-corrections is unlikely to change. Validation accuracy, which the training loop measures with the unigram corrector, exceeds the test accuracy of that same corrector by about four points for every configuration; the validation writers are used for checkpoint selection and are only 55 in number, so this gap is expected and does not indicate a fault. The external systems in Table 5 were trained and evaluated by their authors with their own preprocessing and, in the lower block, on a different partition; we therefore treat only our own runs as a controlled comparison. The extended lexicon is a general English word

list and contains no IAM test transcriptions, but it is a closed lexicon nonetheless, and CRNN-LX should be read as a lexicon-assisted result. Two further caveats concern the trigram. Its corpus is external to IAM; we measured the verbatim overlap with the test lines (Section 3.5) and found idioms only, but a corpus with the same content as the prompts would inflate the result, and any replacement corpus should be checked the same way. And its context is the recognizer’s output for the neighbouring crops, which exists because IAM word images are cut from lines; on a collection of genuinely isolated words the applicable figure is the no-context row of Table 5 (80.74%), not 81.95%. The edit penalty, the decoder and its options were selected on the validation writers and applied once to the test set.

7. Conclusion

CRNN-LX recognizes handwritten words with 81.95% accuracy (8.54% character error rate) on the complete Aachen writer-disjoint test set of IAM, with a plain CRNN trained from scratch, no external image data, and a post-corrector whose only external resource is a public text corpus. Under the same partition and lexicon-free decoding the optical model is 2.6 pp more accurate than the sequence-to-sequence recognizer of Sueiras et al.;²³ the complete system is 0.6–2.7 pp below the later attention-based systems listed in Table 5, and 1.8–3.9 pp below them when the words are treated as isolated. Under controlled conditions, elastic deformation, morphological perturbation and a widened photometric range produce no statistically detectable change in these figures — over three seeds their apparent effect changes sign and averages $+0.20 \pm 0.64$ pp — while removing the conventional augmentation pipeline beneath them costs 2.74 pp, a pattern consistent with the benefit of augmentation saturating before the transforms we proposed; a widely used elastic parameterization, moreover, does not deform the image at all. The post-correction stage, by contrast, moves accuracy by two to three points in either direction, and its four ingredients do different jobs: the lexicon’s coverage of the test vocabulary decides whether the correction helps or hurts; the frequency prior that ranks the candidates supplies most of the gain once coverage is sufficient; an external corpus improves that ranking further; and the line context, read from the recognizer’s own outputs on both sides of a word by decoding the line jointly, adds the last 0.6 pp and returns the character error rate to the lexicon-free level.

For word-level HTR the practical conclusions are that augmentation gains should be demonstrated, not assumed, on the full training set; that an ablation reporting no effect should include the configuration in which the effect is known to exist, and should report the run-to-run spread of an unchanged configuration, which at this scale exceeded every augmentation effect we measured; that lexicon coverage, the ranking rule, its corpus and its context should be reported separately alongside lexicon-assisted accuracy; and that word accuracy and character error rate should be given together whenever a whole-word corrector is used. Future work will test whether the augmentation becomes useful in the low-data regime,

replace the whole-word corrector with a character-level open-vocabulary model to address the uncovered 6% of tokens, and decode the line over the optical model's full output distributions rather than over its single-best hypotheses, so that the language model can also weigh alternatives the greedy decoder discarded.

Acknowledgments

We thank the maintainers of the IAM Handwriting Database and of the Aachen split.

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